

ANGUS (SHENG-CHUN) CHANG

Ithaca, NY 14850 — +1 (978) 530-8356 — angus23232323@gmail.com —
Portfolio: <https://cornell-mae-ug.github.io/fa25-portfolio-ShengChunChang/>

Summary

Mechanical engineering student focused on end-to-end product development, from problem framing to CAD, prototyping. Skilled in hands-on 3D printing and CNC experience, and test planning to close the loop between analysis and hardware. Comfortable with GDT, DFM/DFA, and BOM creation. Proficient in MATLAB and Python for analysis and test automation.

Education

Cornell University, Ithaca, NY
B.S. Mechanical Engineering

Expected May 2027

Skills

- **CAD:** SolidWorks, Autodesk Fusion 360, Onshape, ANSYS
- **Programming:** Arduino, Python, Java, C++, MATLAB
- **UAV Systems:** FPV drone design/building, Betaflight tuning
- **Certifications:** Remote Pilot (FAA Part 107), Red Cross Lifeguard/CPR/AED

Research & Patents

- An “Artificial Leaf” generating oxygen and electricity — Patented in Taiwan and Germany
- Fabric structure utilizing protein denaturation and adhesion — Patented in Taiwan
- Tunable reflectance based on phase change metamaterial for structural color filter

Experience

Autonomous Drone Project Team (CUAD)

Sep 2025 – Present

Mechanical Subteam Member

- Balance lightweight structures with aerodynamic efficiency for UAV performance
- Design and manufactured drone frames/payloads using 3D printing and CNC machining

Bio-Inspired Fluid Lab, Cornell University

Sep 2025 – Present

Undergraduate Researcher — Bat Robot

- Assist with CAD modeling, lightweight wing structure fabrication, and aerodynamic optimization to replicate bat flight mechanics.

Cornell University

Sep 2024 – Jan 2025

Undergraduate Researcher — Clustering Greenland Seismological Data

- Cluster earthquake data from Greenland using AI machine learning

Achievements

- Top 300 Regeneron Science Talent Search Scholar, 2023
- ISEF 1st Place Chemistry Special Award, 2022
- Columbia Business School Venture for All Model Entrepreneur Competition, 1st Place, 2023
- Published in *International Journal of High School Research*, Vol. 5, Issue 3